

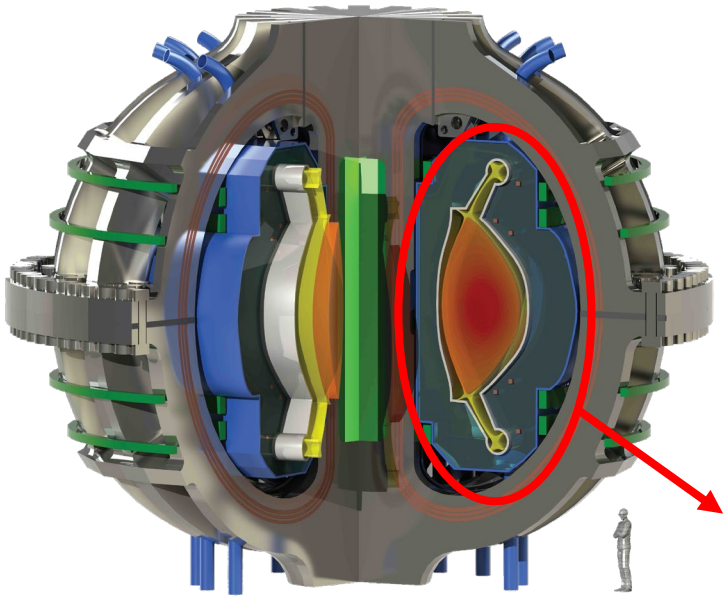
Neutronics of the LIBRA Project

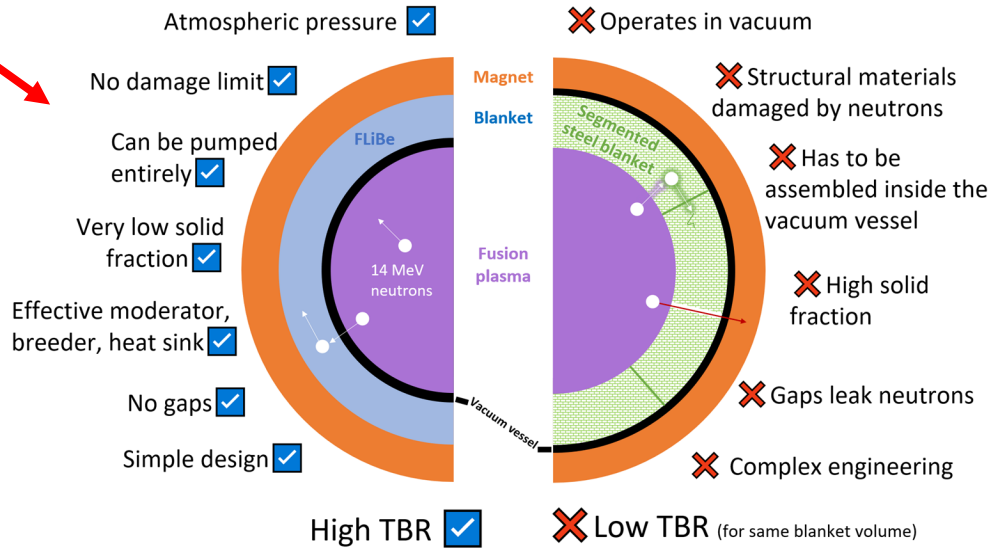
Stefano Segantin
& LIBRA team
Postdoctoral Associate, PSFC-MIT

MIT PSFC LIBRA

Liquid immersion blanket is simpler

- Almost no structure
- Low p, high T
- Reduced radiation damage
- High TBR
- Multi-purpose fluid:
  - Breeder
  - Moderator/Shield
  - Heat sink





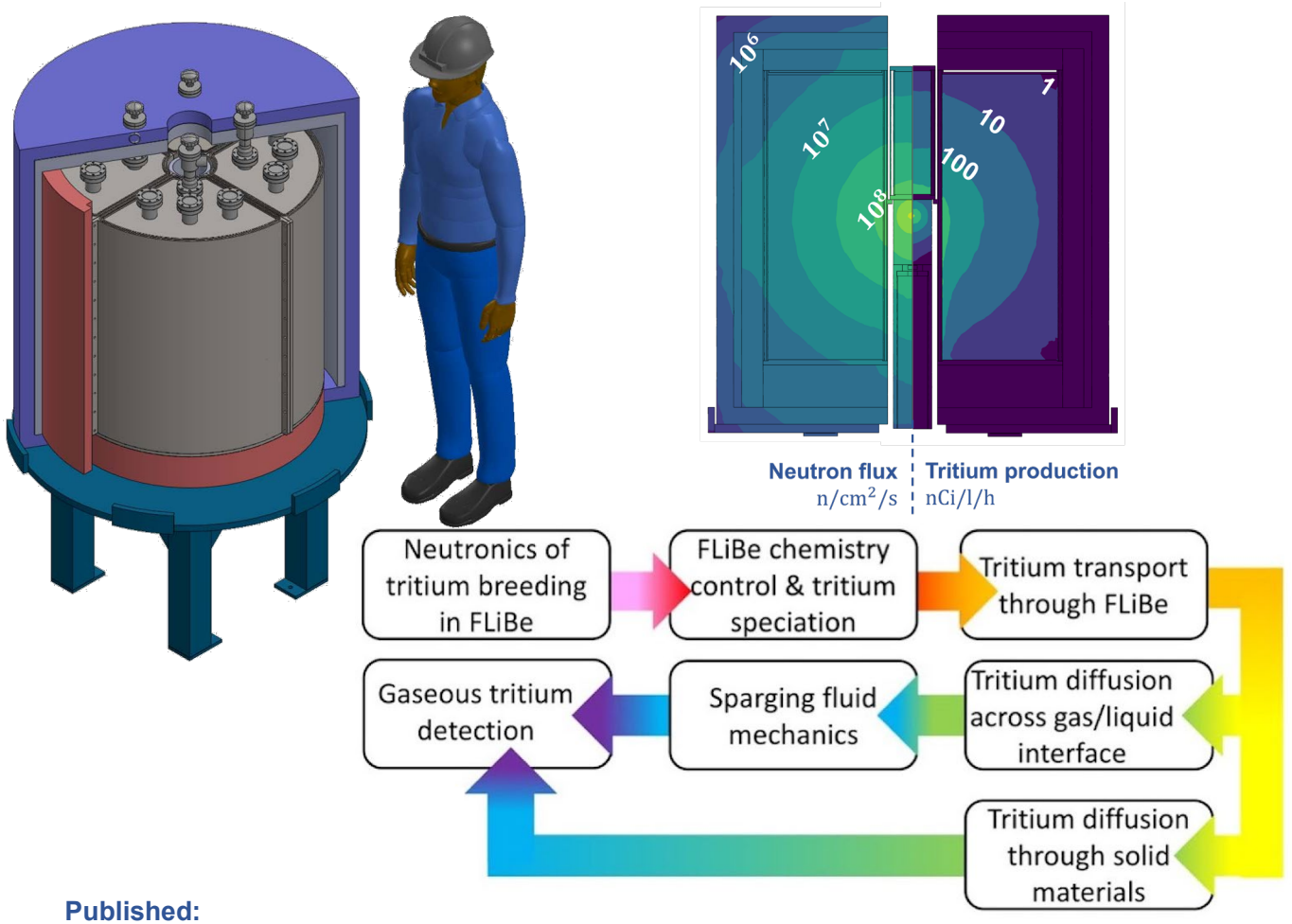
Liquid Immersion Blanket Robust Accountancy

Objectives:

- **Breed tritium** with 14 MeV fusion neutrons
- **Demonstrate LIB concept** at fusion relevant conditions (high T, molten salt breeder, fusion neutrons, fusion materials etc.)
- Develop and test relevant **equipment** and SOPs
- **TBR ~ 1**

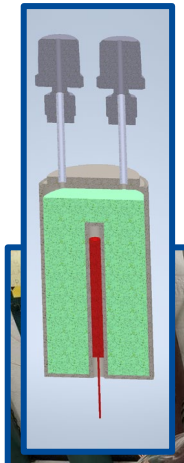
Challenges:

- **Hazard management** (high T, **radiation**, beryllium, fluorine etc.)
- **Tritium accountancy**
- **TBR measurement** (neutron accountancy)



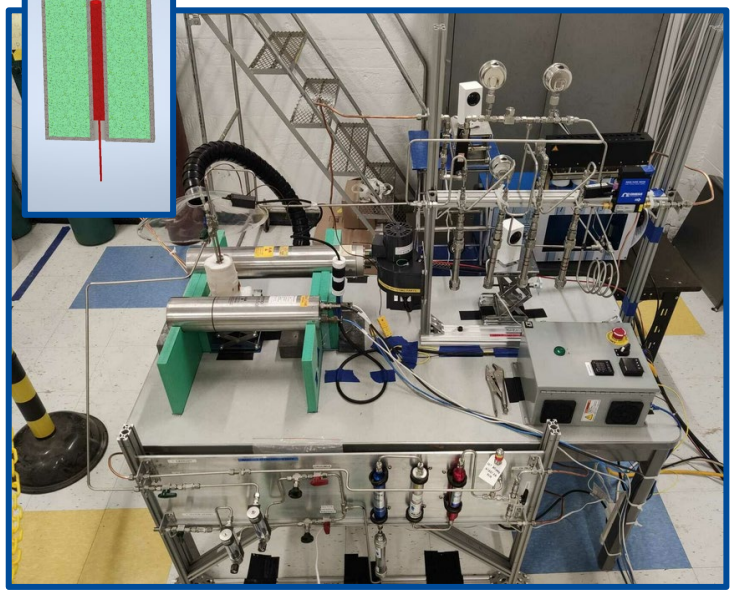
Published:
Ferry S. E. et al. Fusion Science and Technology (2023).

BABY to LIBRA: we are scaling up

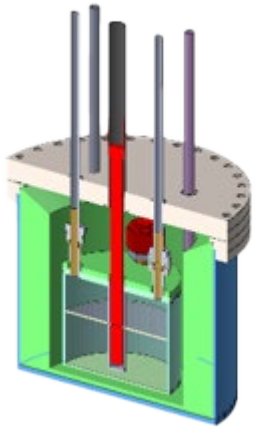


100 ml BABY

ClLiF salt



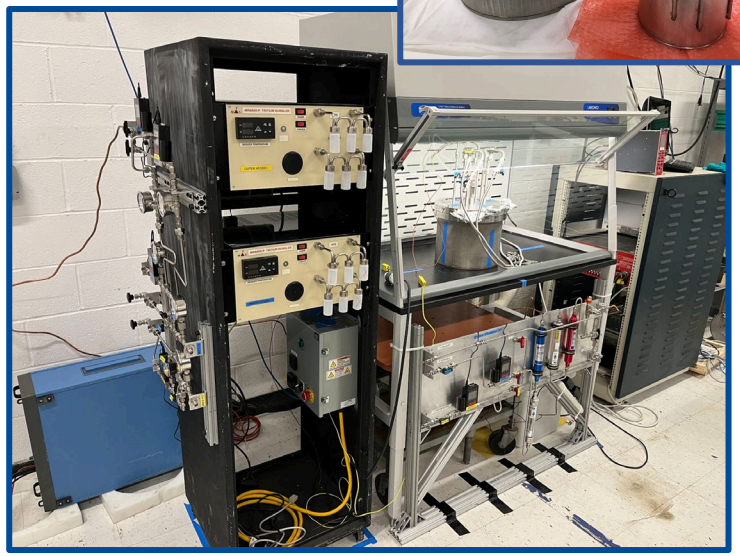
Done



1 liter BABY

ClLiF & FLiBe



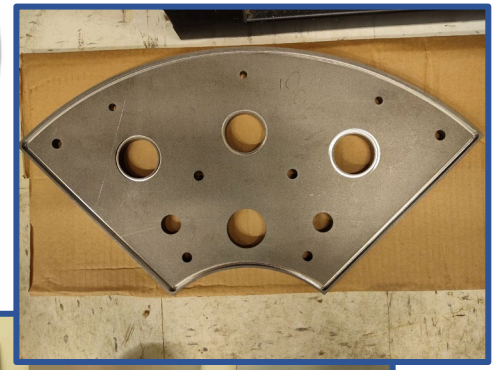


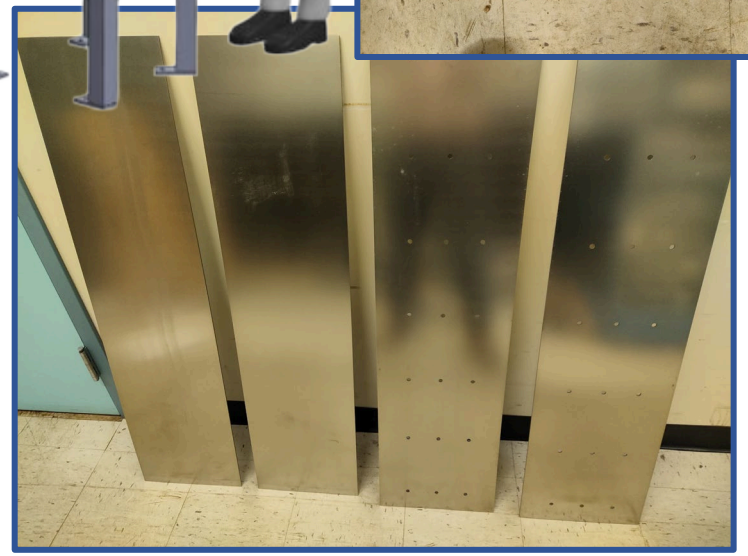
Running



LIBRA Pi

ClLiF & FLiBe





In construction

MIT PSFC LIBRA

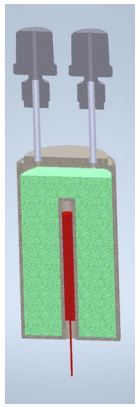
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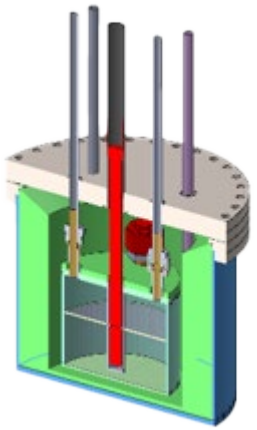
LIBRA: multi-disciplinary activities and research

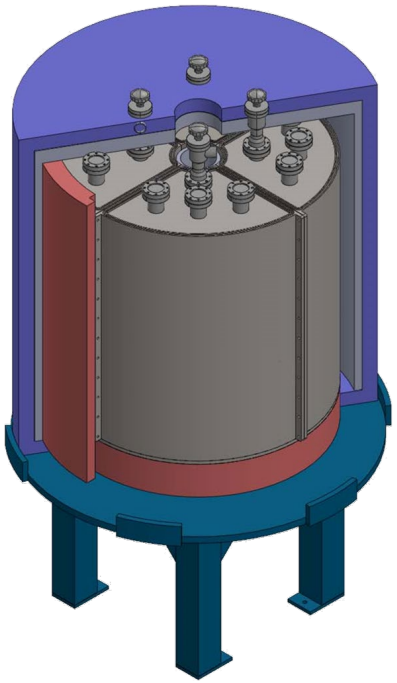
Neutronics

- Exploration of breeding compounds
- Neutron detection
- Experiment design
- Test advanced neutronics workflows

LIB experiments







Measured TBR

Tritium

- Transport Modeling
- Collection and detection
- Speciation

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Identified ClLiF as good surrogate salt for LIBRA (and LIB)

Neutronics

Exploration of breeding compounds

Neutron detection

Experiment design

Test advanced neutronics workflows

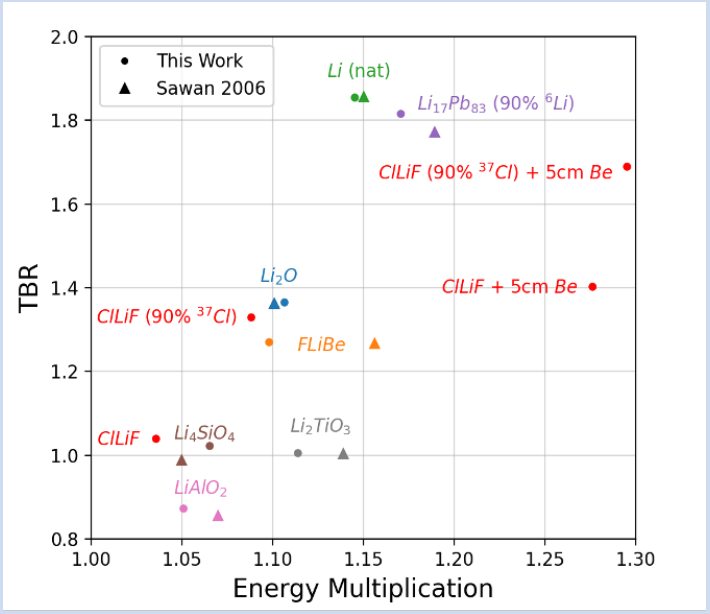
LIB experiments

Measured TBR



Student Spotlight: Collin Dunn

TOFE talk: *Neutronics Analysis and Uncertainty Quantification of LiCl-LiF Molten Salt as a Liquid Tritium Breeding Blanket*
Monday, July 22, 2024 | 3:15–5:05PM CDT
cdunn314@mit.edu



Tritium

Transport Modeling

Collection and detection

Speciation

Tritium detection is the first pillar of a robust TBR measurement

Neutronics

- Exploration of breeding compounds
- Neutron detection
- Experiment design
- Test advanced neutronics workflows

LIB experiments

Measured TBR

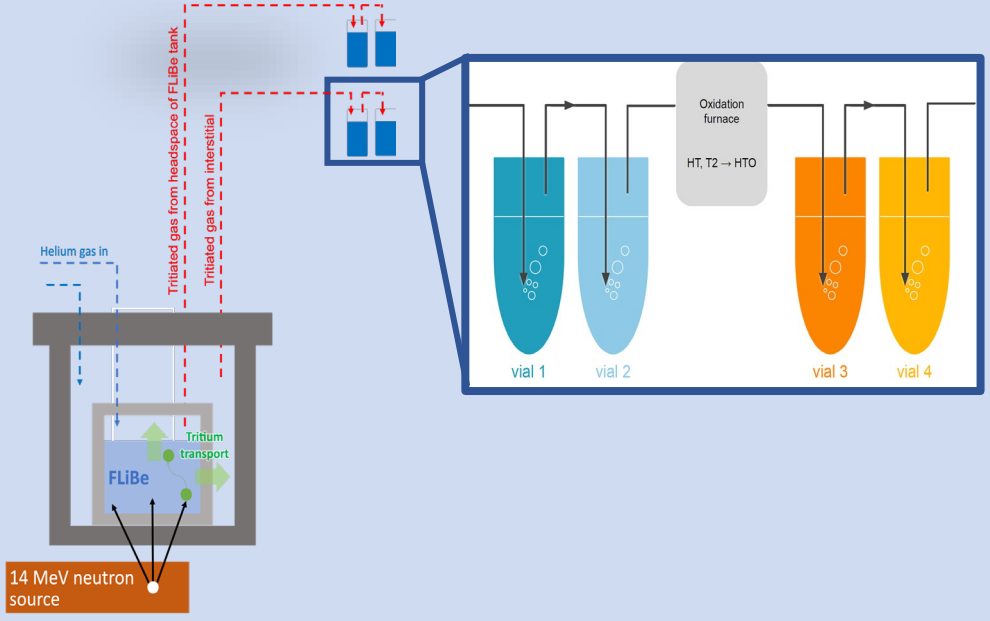


Student Spotlight: Nikola Goles

TOFE talk: *Tritium Speciation and Transport in Molten Salts: Insights from the BABY Experiment*

Monday, July 22, 2024|3:15–5:05PM CDT

goles@mit.edu



Tritium

- Transport Modeling
- Collection and detection
- Speciation

Neutron detection completes the measured TBR

Neutronics

- Exploration of breeding compounds
- Neutron detection
- Experiment design
- Test advanced neutronics workflows

LIB experiments

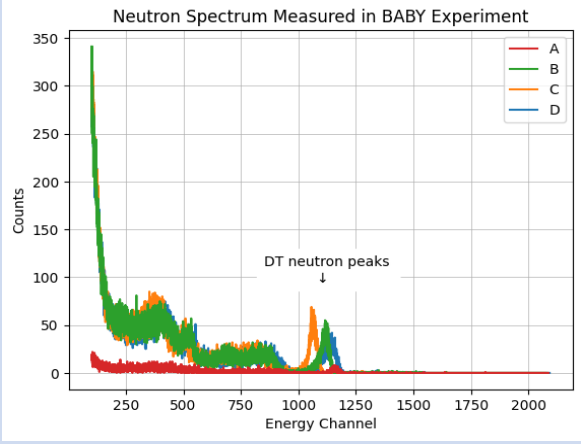
Measured TBR

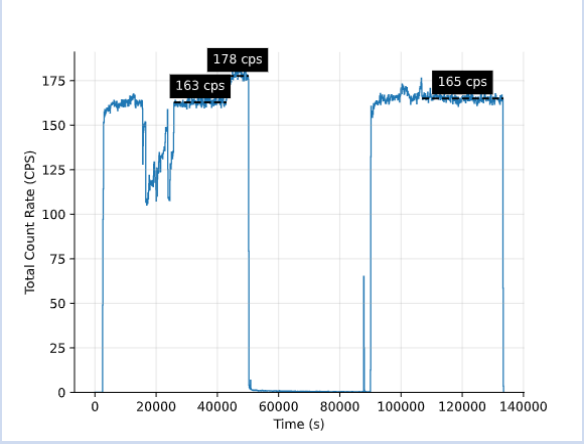


Student Spotlight: Emily Edwards

Calibration of proton recoil detector
DT compact neutron source characterization

emily424@mit.edu





Tritium

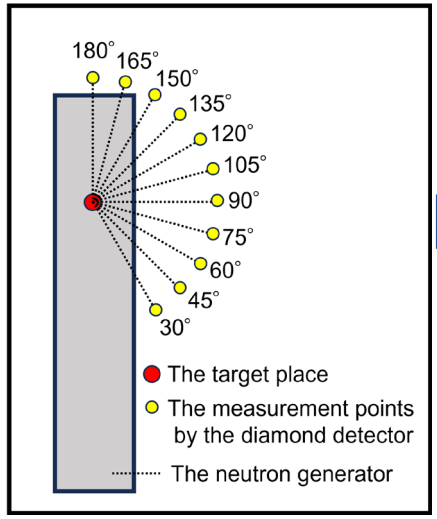
- Transport Modeling
- Collection and detection
- Speciation

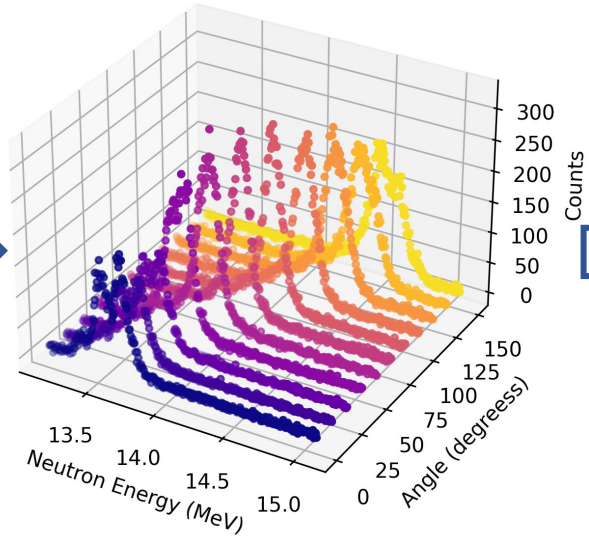
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14 MeV neutron source characterized

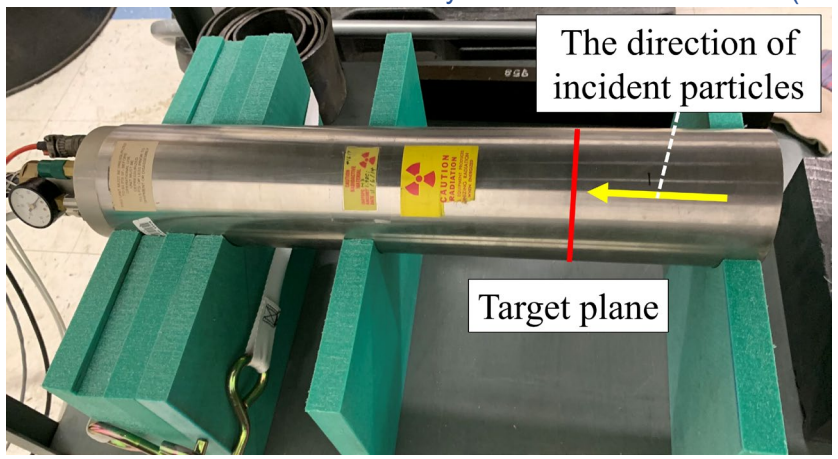
- Diamond detector
- Activation foils
- Energy-angular distribution implemented in OpenMC neutronics models





OpenMC

Courtesy of J.L. Ball and S. Mackie (MIT PSFC)



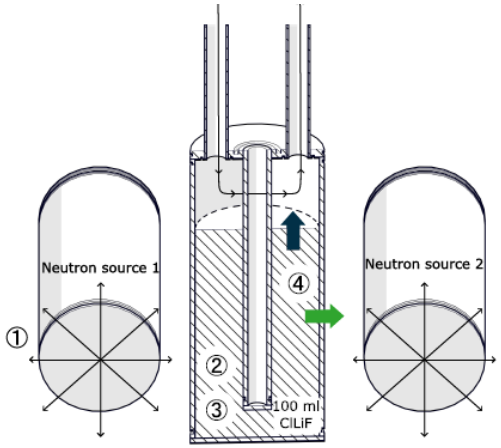
Published:
Sakabe T. et al. IEEE Transactions on Plasma Science (2024).

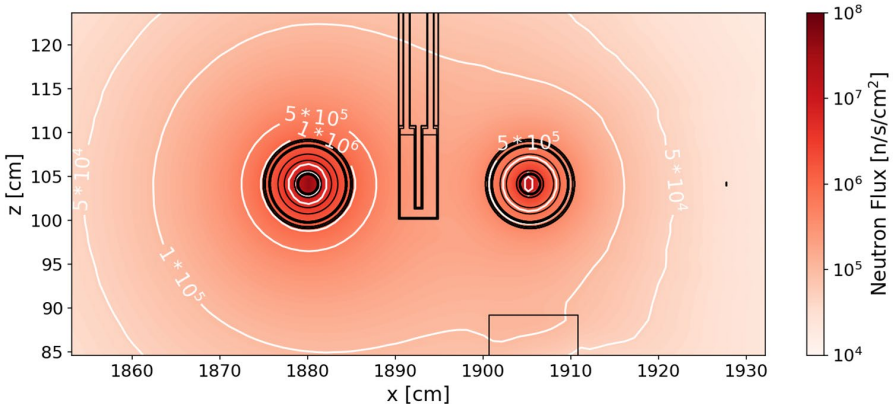
MIT PSFC LIBRA

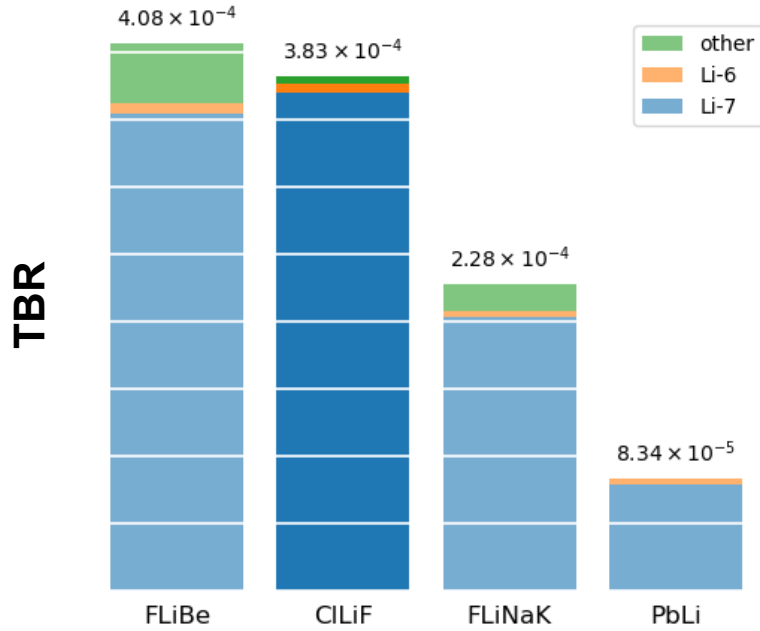
ANS

Experiment design: 100 ml ClLiF focus

- Model geometry and materials
- Compute **expected TBR**
- **Shielding design** and radiation protection assessment
- Test different **breeding compounds** at given configuration





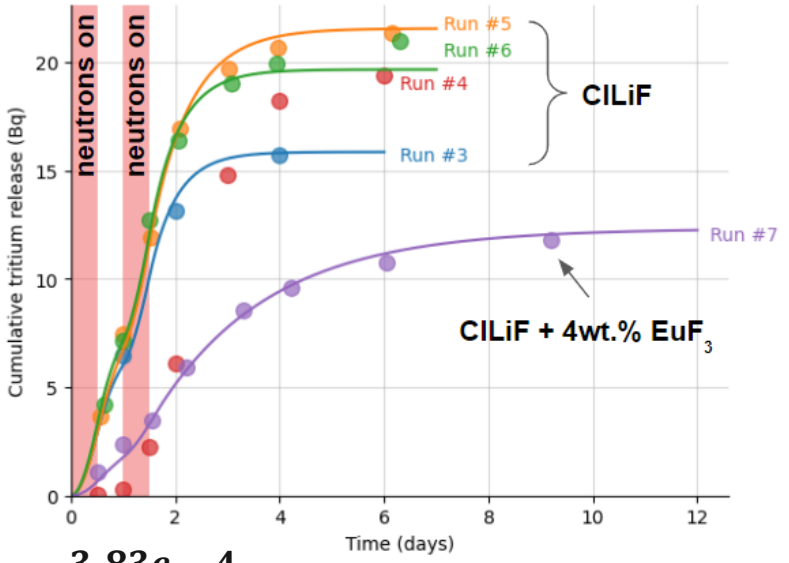


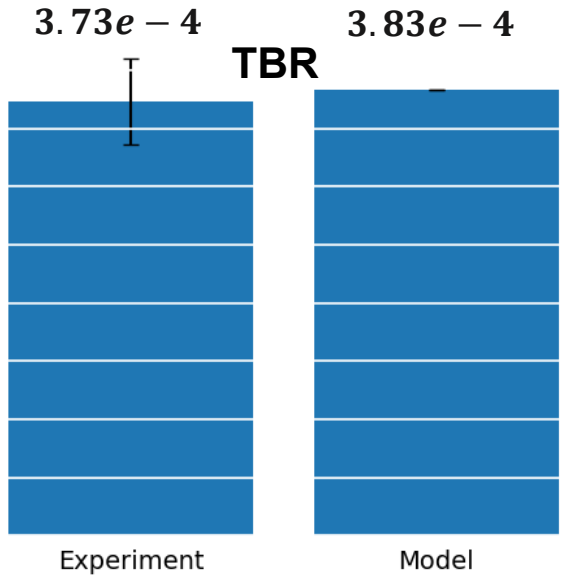
100 ml BABY - CILiF bred tritium consistently

- **Bred ~ 1 Bq/h of T** for 24 hours
- Good **reproducibility**
- Provided insights on salt chemistry (run 7)

Challenges/uncertainties:

- **T-wall permeation** →1 liter BABY
- Salt characterization (**density**)
- Neutron detection and **source characterization**





Submitted:
Delaporte-Mathurin R. et al. Nuclear Fusion (2024).

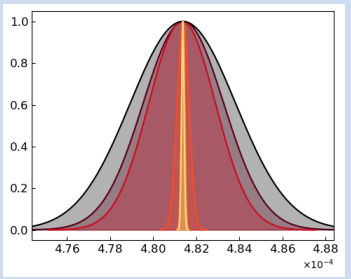
Advanced neutronics workflows for validation and robust accountancy

Sources of uncertainty on models

OpenMC

(https://github.com/openmc-dev/openmc)

Computed TBR



Sources of uncertainty on measurements

$$\frac{TPR_{exp} \pm \sigma}{source\ rate_{exp} \pm \sigma}$$

Measured TBR ± uncertainty (σ)

MIT PSFC LIBRA

ANS

Advanced neutronics workflows for validation and robust accountancy: models UQ

Sources of uncertainty on models

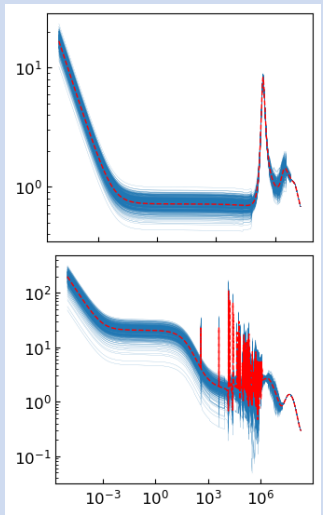
Nuclear data

DT source characteristics

Salt density as f(T)

...

Perturbed xs (implemented)

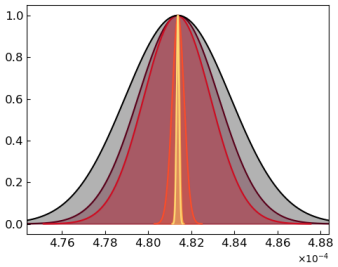


...

Sandy (github.com/luca-fiorito-11/sandy)

OpenMC

Computed TBR



$$\frac{TPR_{exp} \pm U}{source\ rate_{exp} \pm U}$$

Measured TBR ± uncertainty

Ebiwonjumi B.
TOFE talk: *Uncertainty Analysis of OpenMC Shutdown Dose Rates Calculation*
Wednesday, July 24, 2024|10:15AM–12:05PM CDT

MIT PSFC LIBRA

ANS

Advanced neutronics workflows for validation and robust accountancy – models UQ

Sources of uncertainty on models

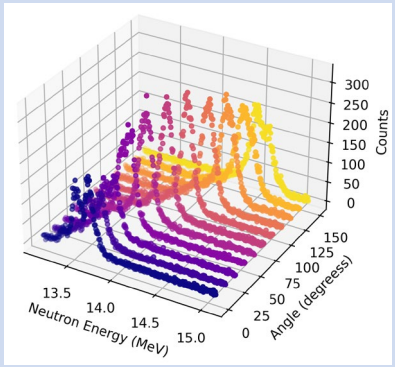
Nuclear data

DT source characteristics

Salt density as f(T)

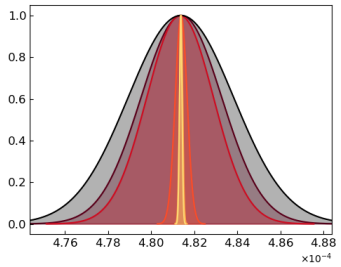
…

Measured distributions (future works)



OpenMC

Computed TBR



$$\frac{TPR_{exp} \pm U}{source\ rate_{exp} \pm U}$$

Measured TBR ± uncertainty

MIT PSFC LIBRA

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Advanced neutronics workflows for validation and robust accountancy: measurements UQ

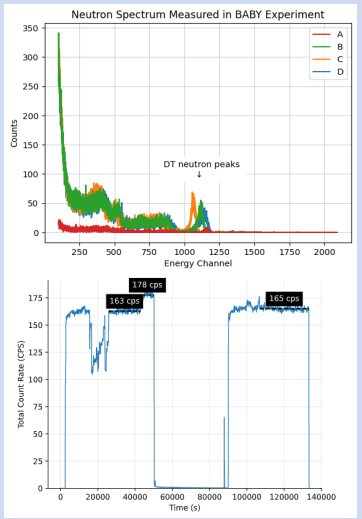
Sources of uncertainty on measurements

DT source rate measurements

T counting on LSC

…

Measured source rate (implemented – to improve)



…

OpenMC

Computed TBR

$$\frac{TPR_{exp} \pm U}{source\ rate_{exp} \pm U}$$

Measured TBR ± uncertainty

ANS

Advanced neutronics workflows for validation and robust accountancy: measurements UQ

Sources of uncertainty on measurements

DT source rate measurements

T counting on LSC

…

Quantify uncertainty on tritium concentration (this summer)





…

OpenMC

Computed TBR

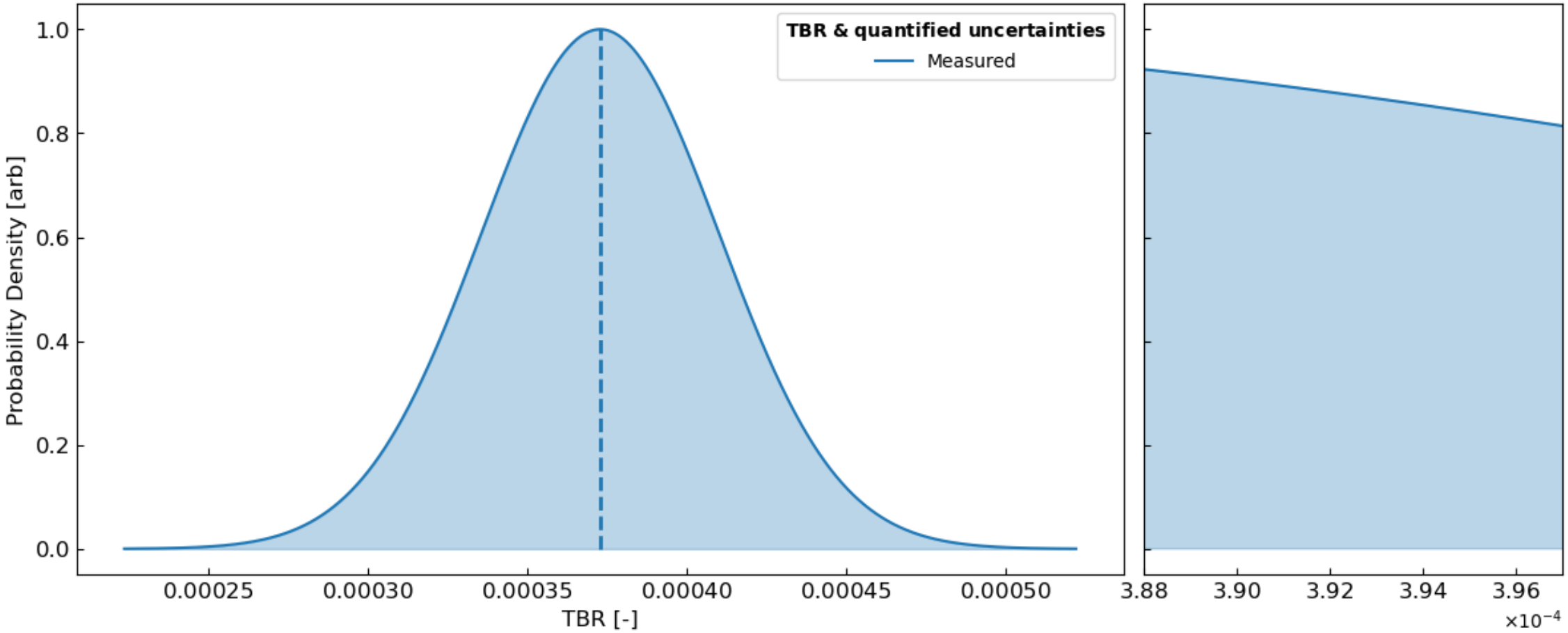
$$\frac{TPR_{exp} \pm U}{source\ rate_{exp} \pm U}$$

Measured TBR ± uncertainty

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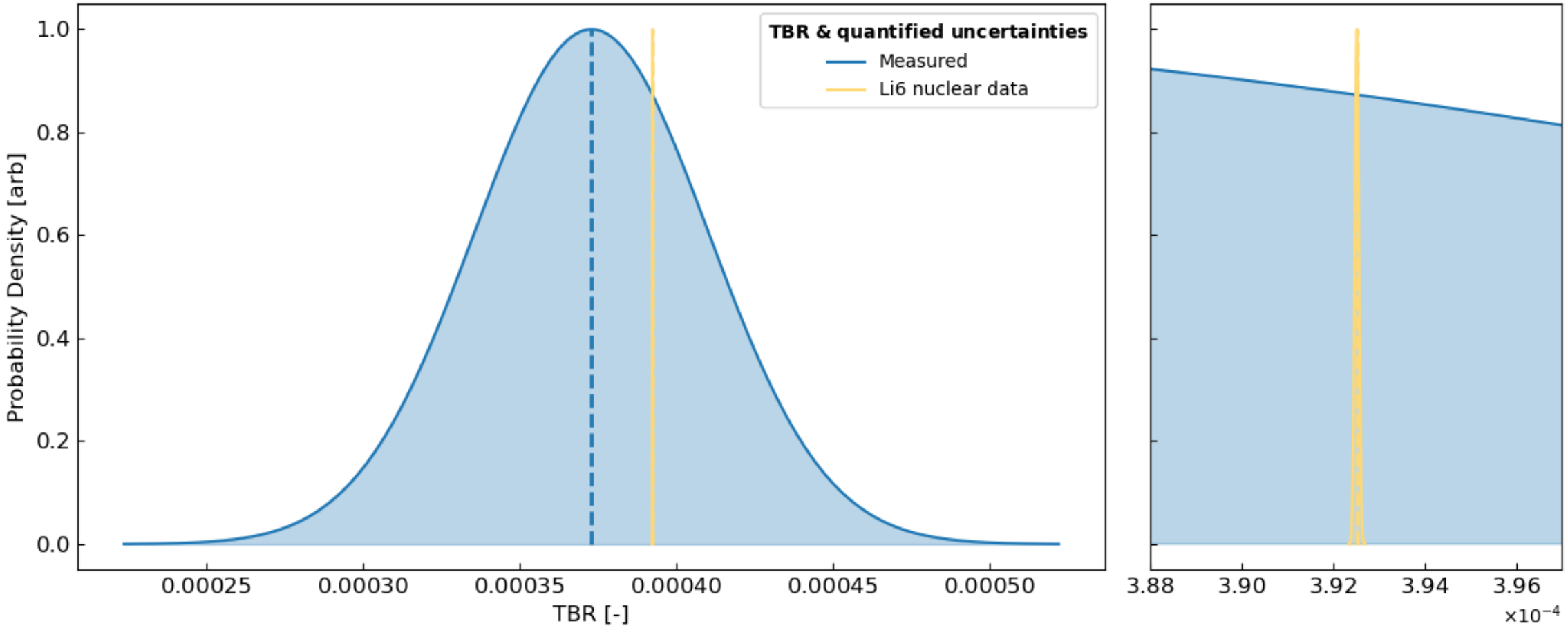
Models and UQ strengthen tritium accountancy reliability



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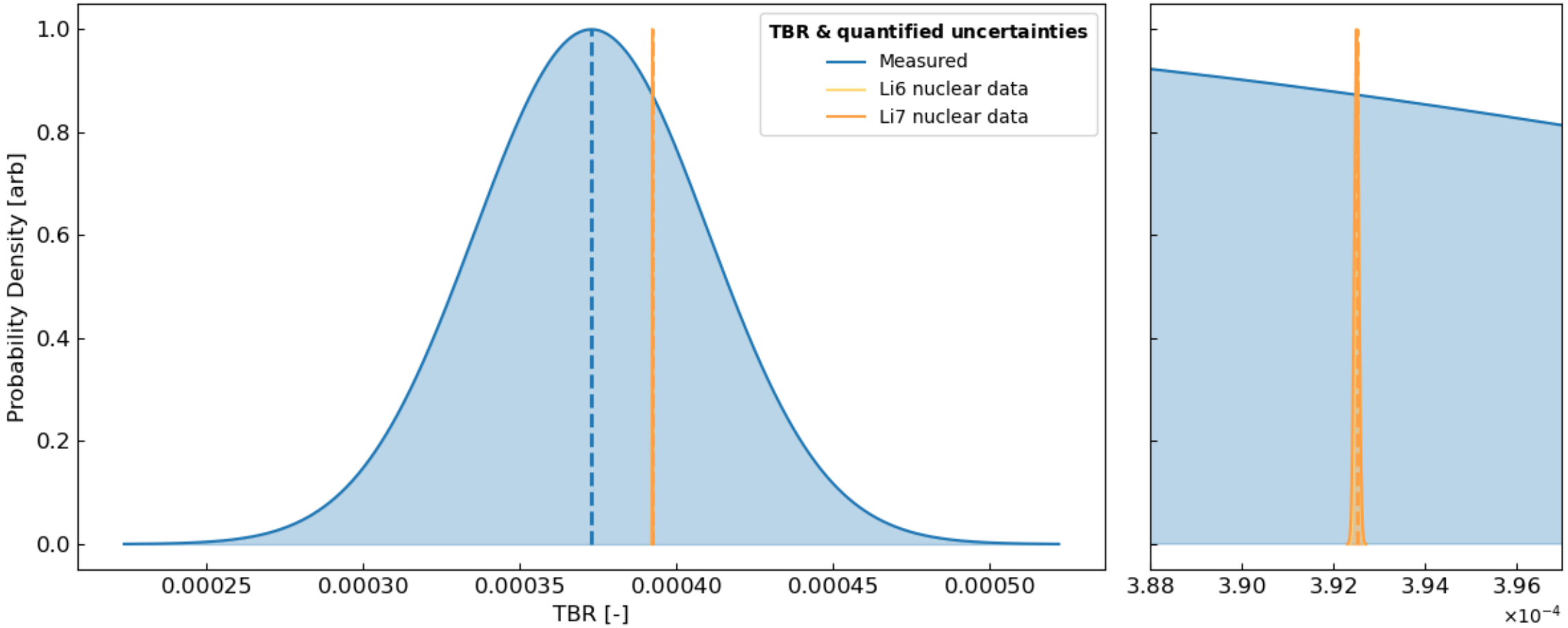
Models and UQ strengthen tritium accountancy reliability



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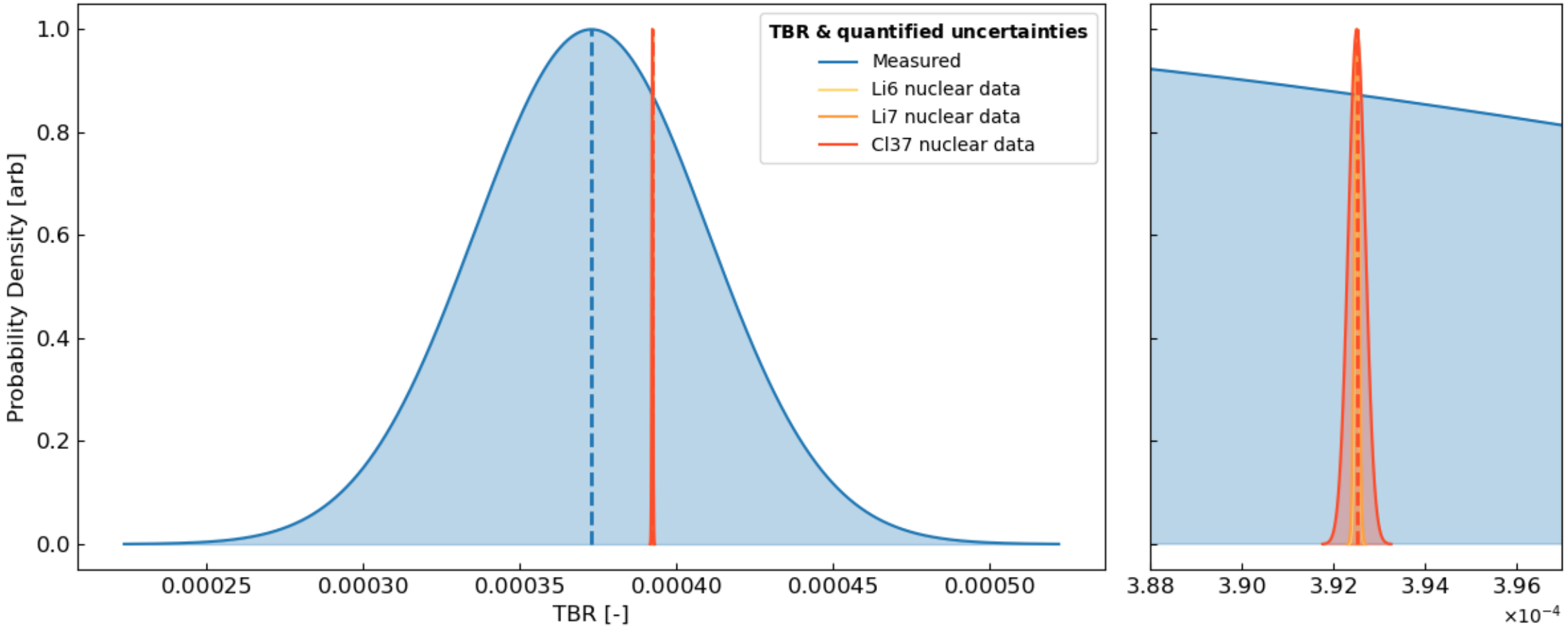
Models and UQ strengthen tritium accountancy reliability



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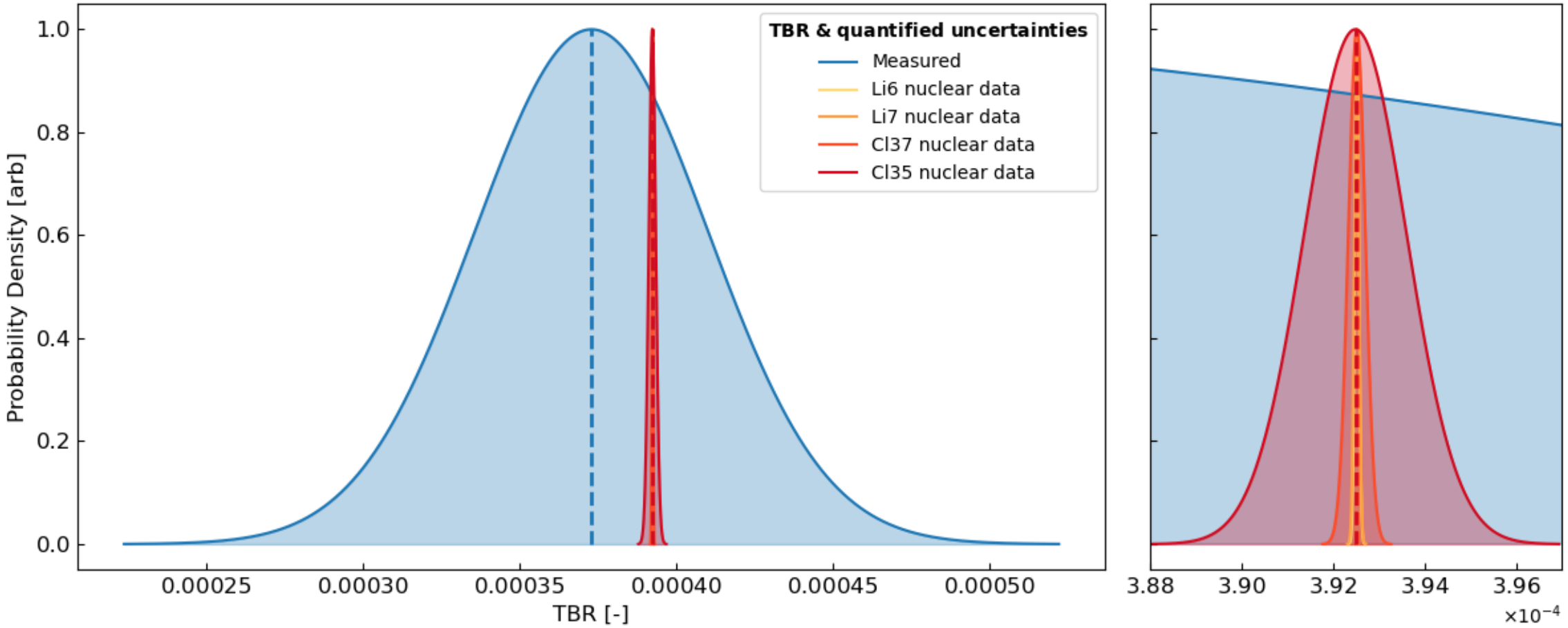
Models and UQ strengthen tritium accountancy reliability



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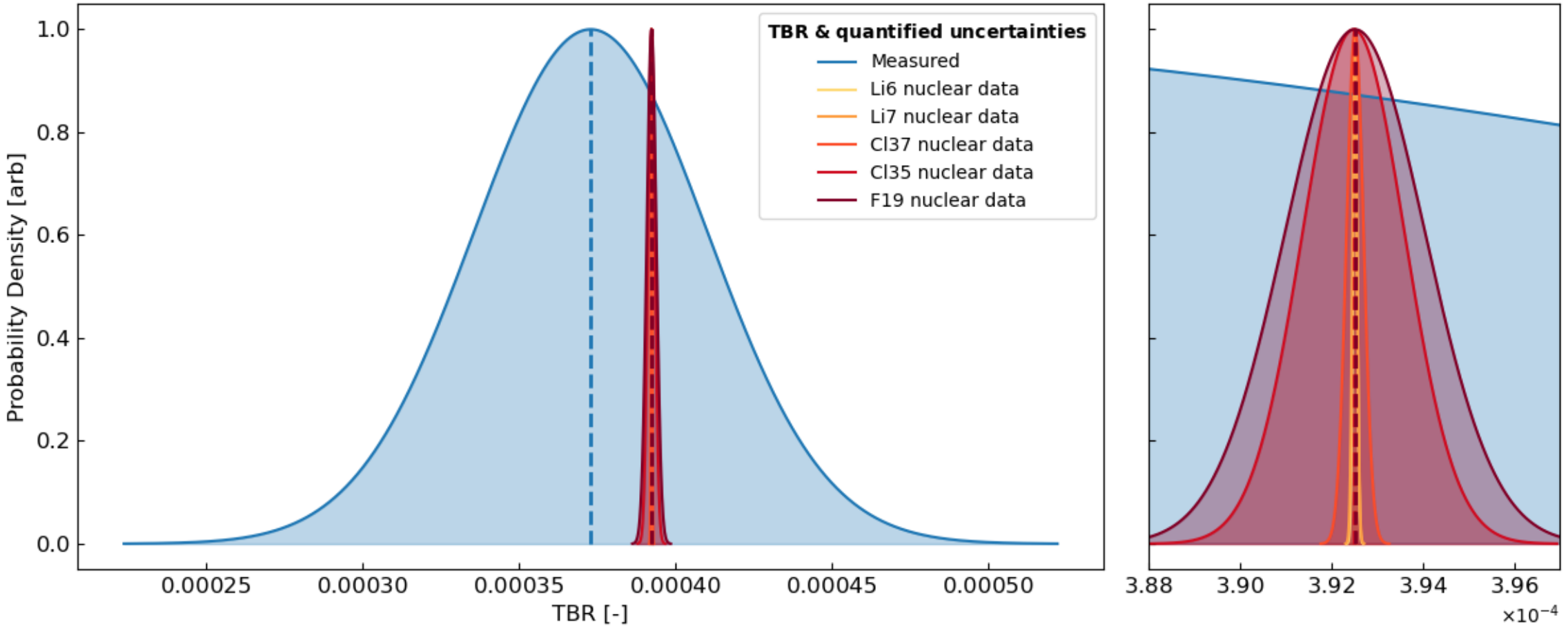
Models and UQ strengthen tritium accountancy reliability



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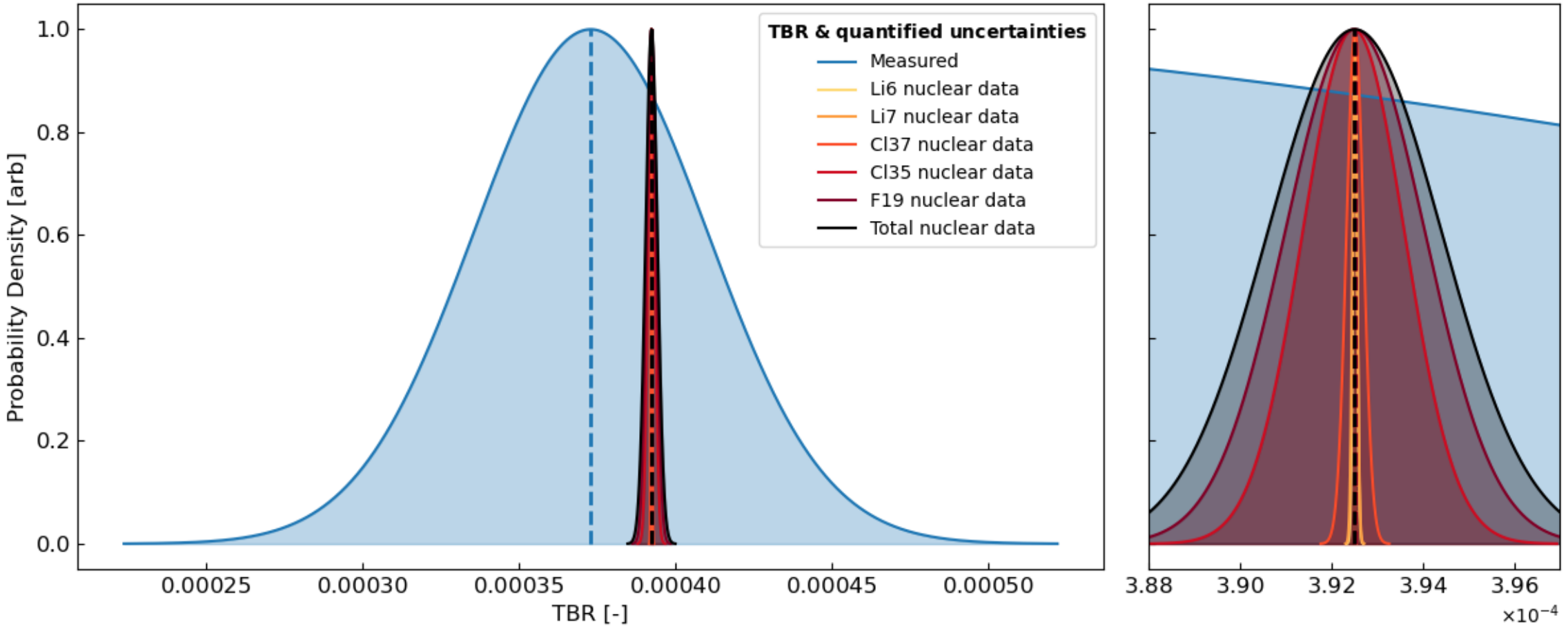
Models and UQ strengthen tritium accountancy reliability



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Models and UQ strengthen tritium accountancy reliability



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Key takeaways

- **LIBRA is a good test bed** for advanced fusion neutronics workflows
- **Design → Measure ←→ Validate** is a relatively simple process to go through in order to cross validate measurements and models/methods
- **100 ml BABY** is a **good setup for accelerated learning** and fast iterations but it is missing an important piece (tritium permeated through crucible walls)
- Action needs to be taken in order to **reduce uncertainty** on both measuring and modeling sides
- **1 liter BABY** needs to tackle such uncertainties

Future works

- **Scaling up to 1 liter BABY** (and following LIBRA designs) we will have access to permeated tritium through wall accountancy – a so far missing piece that is important for neutronics modeling and validation as well
- Implementation of **proton recoil** and **scintillation detectors** for neutron source characteristics
- Implementation of **additional sources of uncertainty** in the TBR UQ algorithm (e.g. neutron generator energy spectra, neutron rate, salt density etc.)
- **Implementation of CAD-unstructured mesh transport** in neutronics workflows

Authors & Acknowledgments

- Segantin S., Dunn C.S., Edwards E., Lanzrath A., Peterson E.E. & LIBRA team
- Panontin E., Ball J.L. & PSFC energetic particles physics team
- Ebiwonjumi B. & PSFC fusion neutronics team
- Vieira R., Leccacorvi R. & PSFC engineering team

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Thank you!

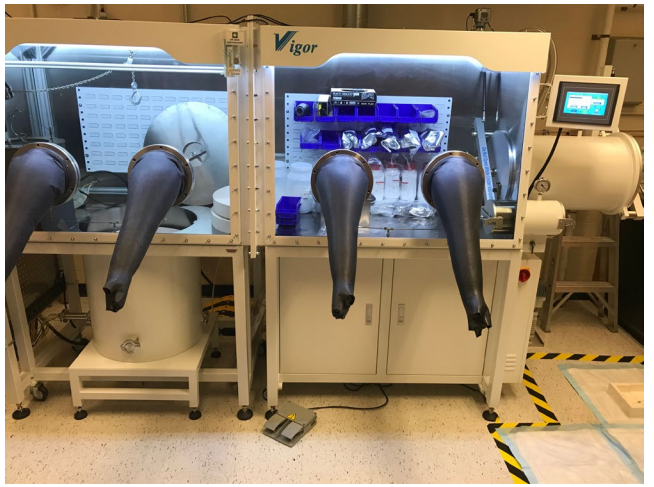
MIT PSFC LIBRA

Backup slides

Salt handling

- ClLiF surrogate salt for experience
  (spoiler alert: ClLiF has potential for blanket as well)
- Glove box handling
- Double containment
- Be sensors

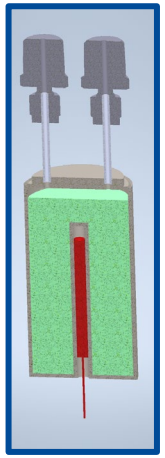


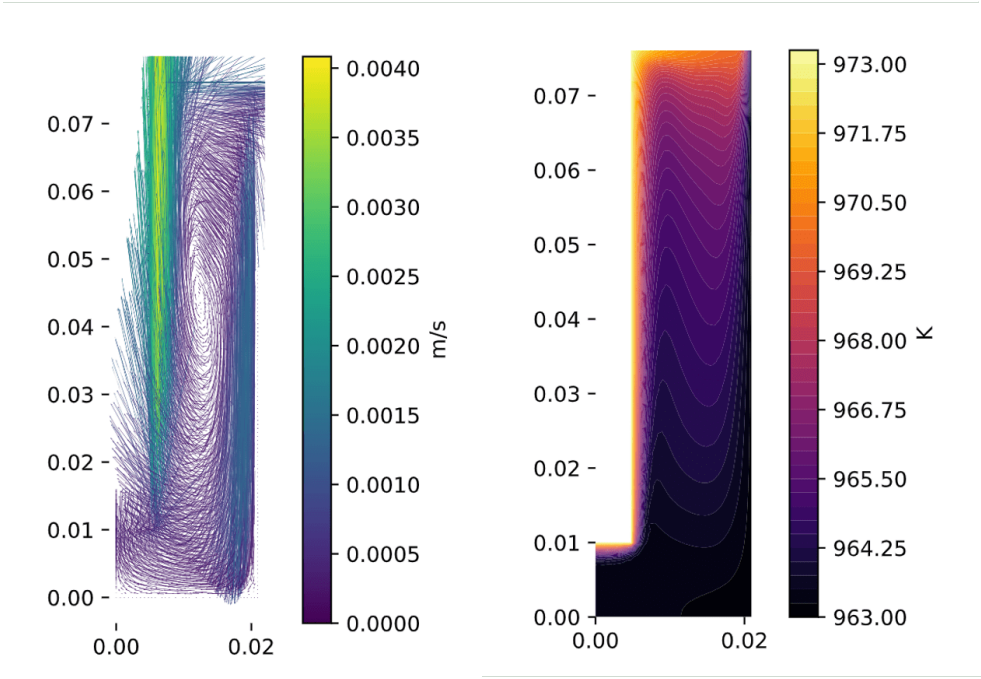




Tritium transport modeling

- Zero-D fast assessment of T-transport dynamics
- Evaluation of transport coefficients
- 2-D and 3-D complex modeling of transport phenomena with FESTIM





courtesy of C. Weaver (MIT)

MIT PSFC LIBRA

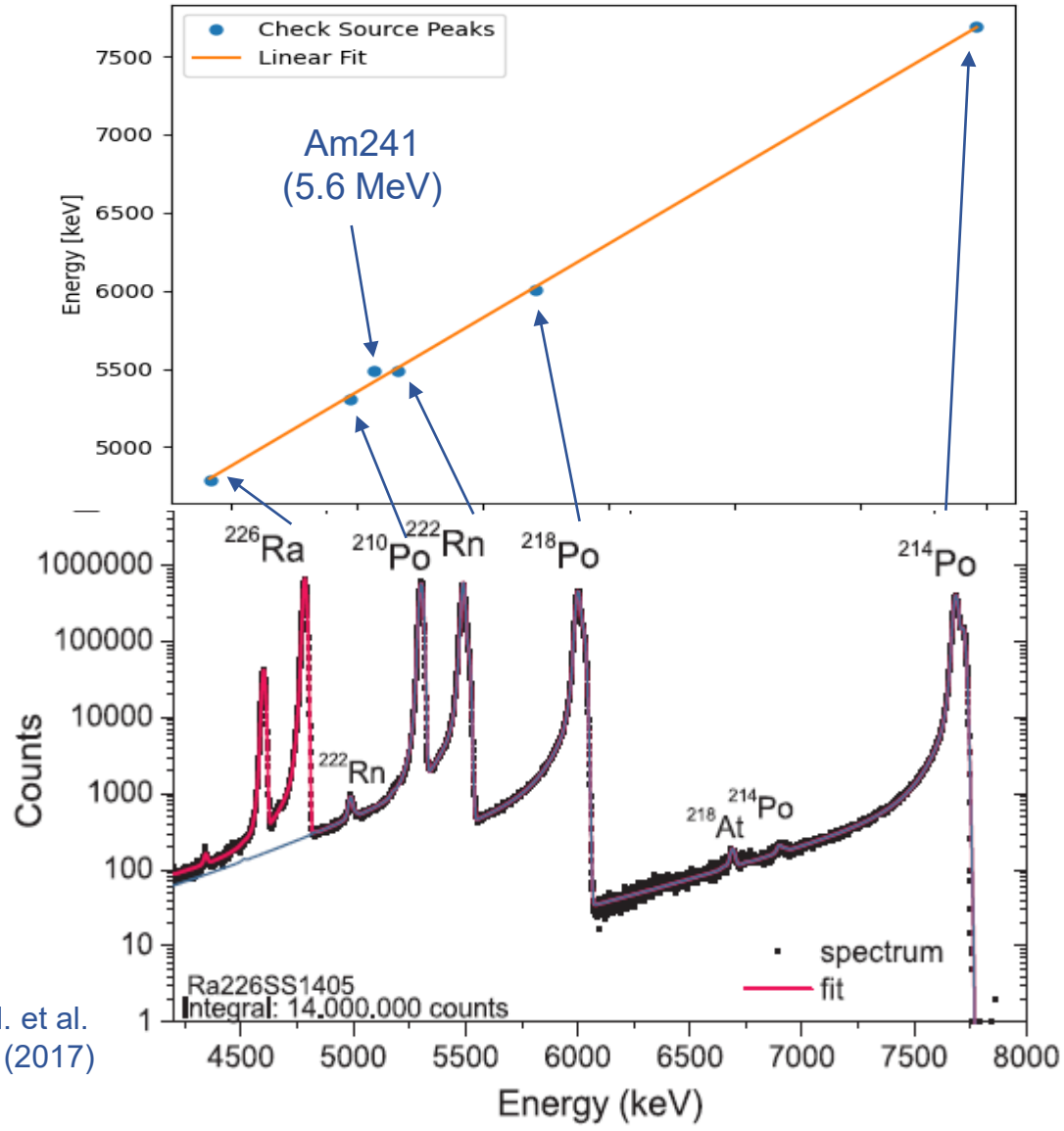
ANS

Diamond detector calibration

Alpha check sources in fast region for calibration

- Am241 (1 peak)
- Ra226 (5 peaks)

Robust workflow: running calibration overnight right before LIBRA rub

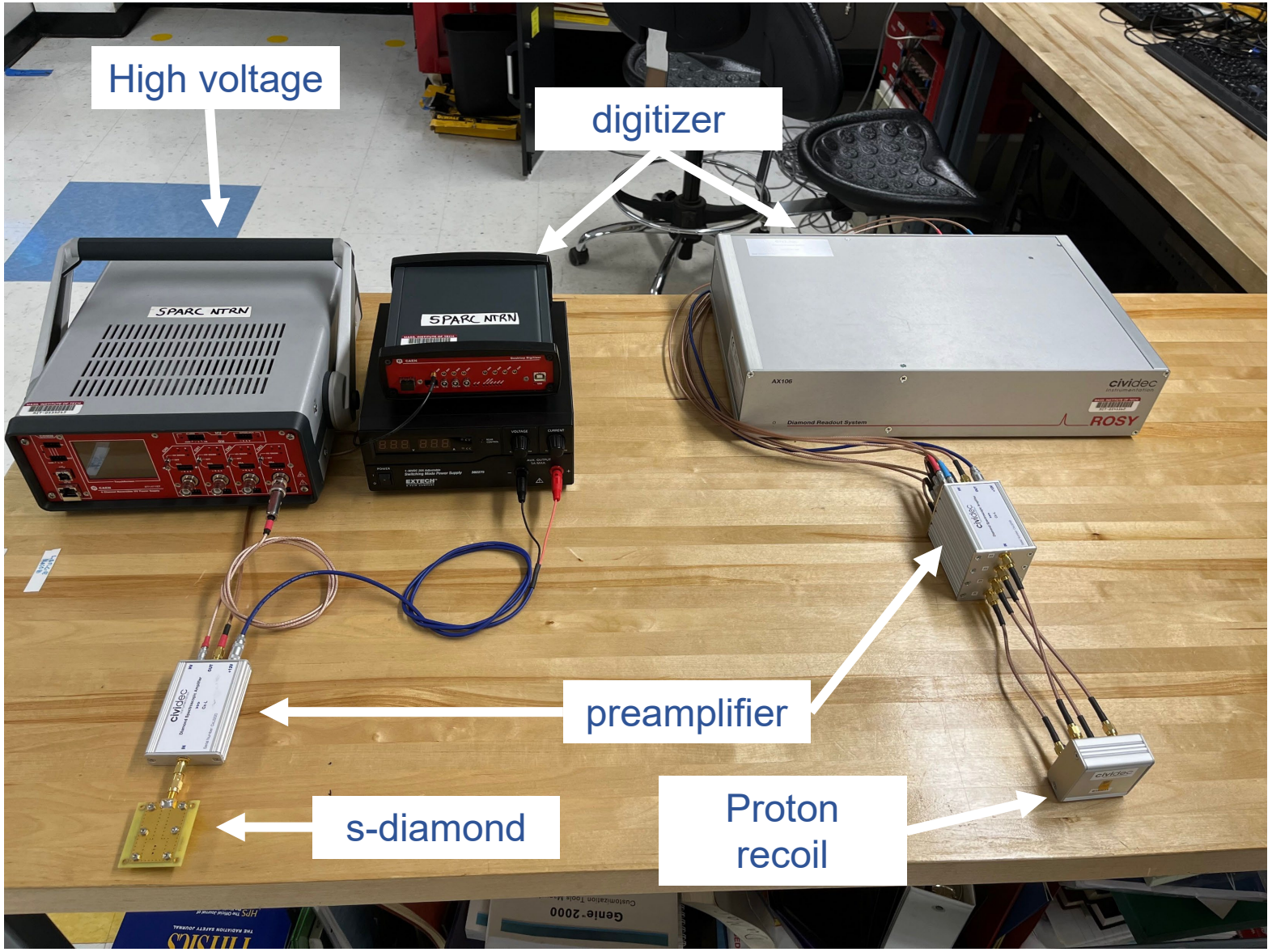


Marouli, M. et al. Ap.Rad.Is (2017)

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Neutron detection



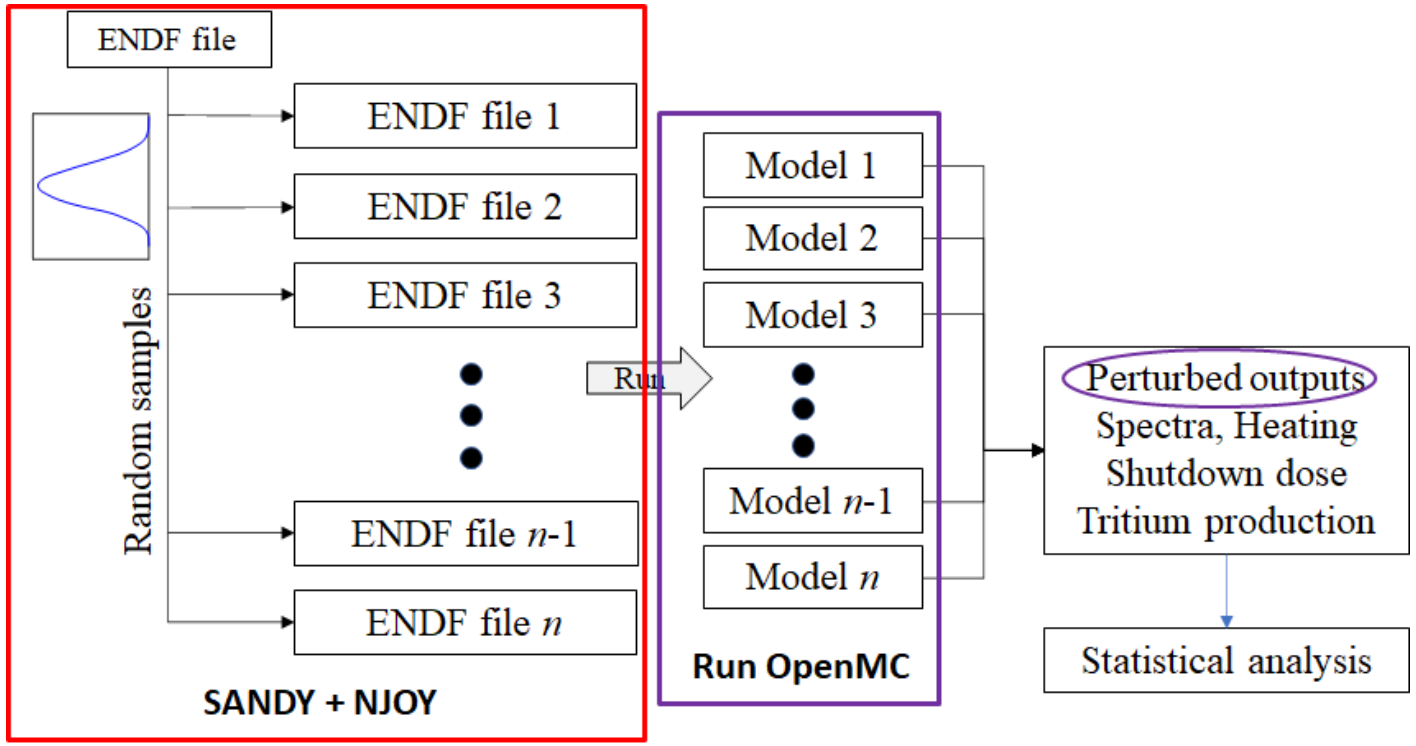


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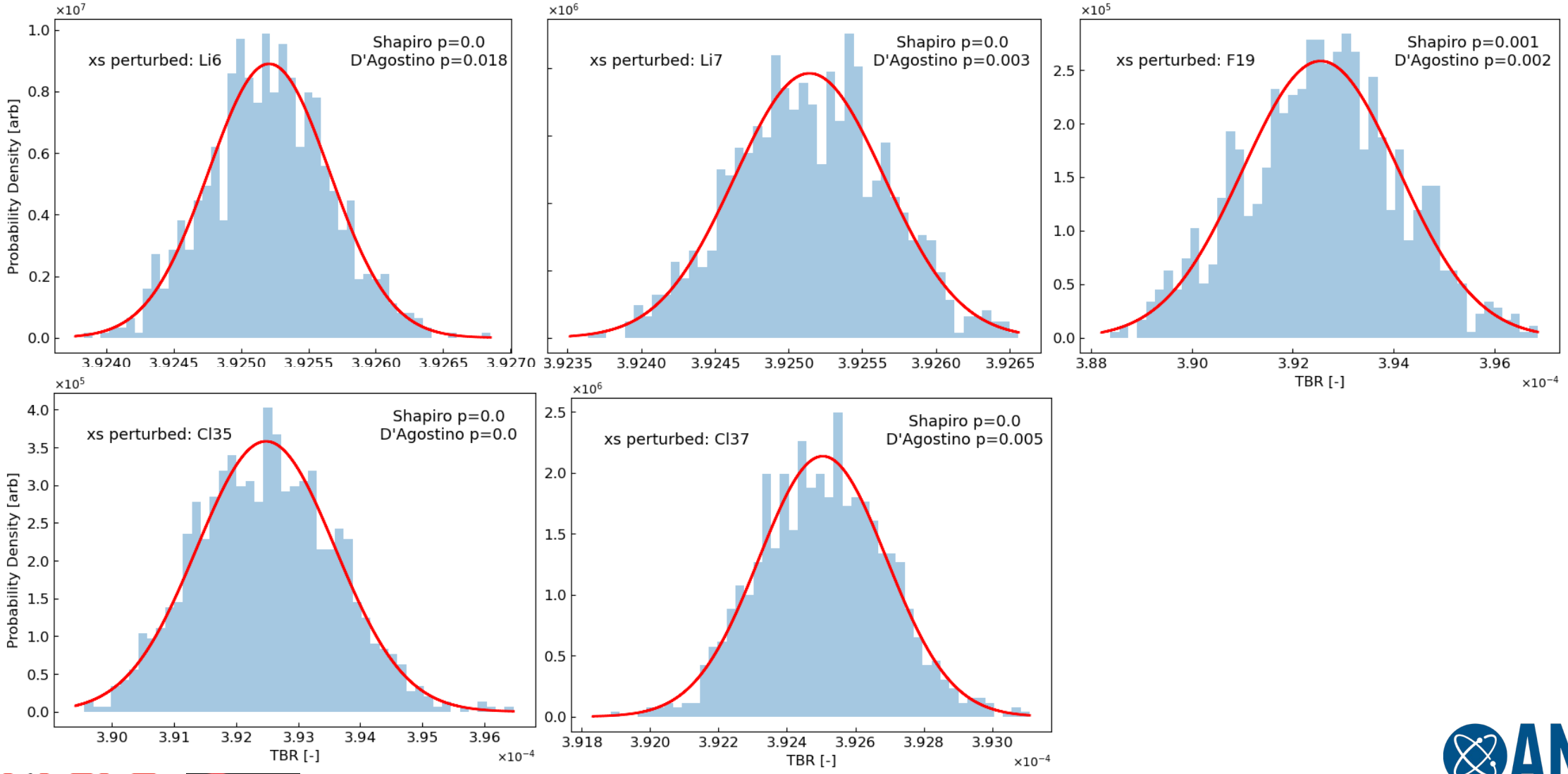
Additional OpenMC UQ methodology

- Total Monte Carlo method used
- 500 simulations per perturbed xs done
- Need to check stabilization through error evolution and relative std of the relative std

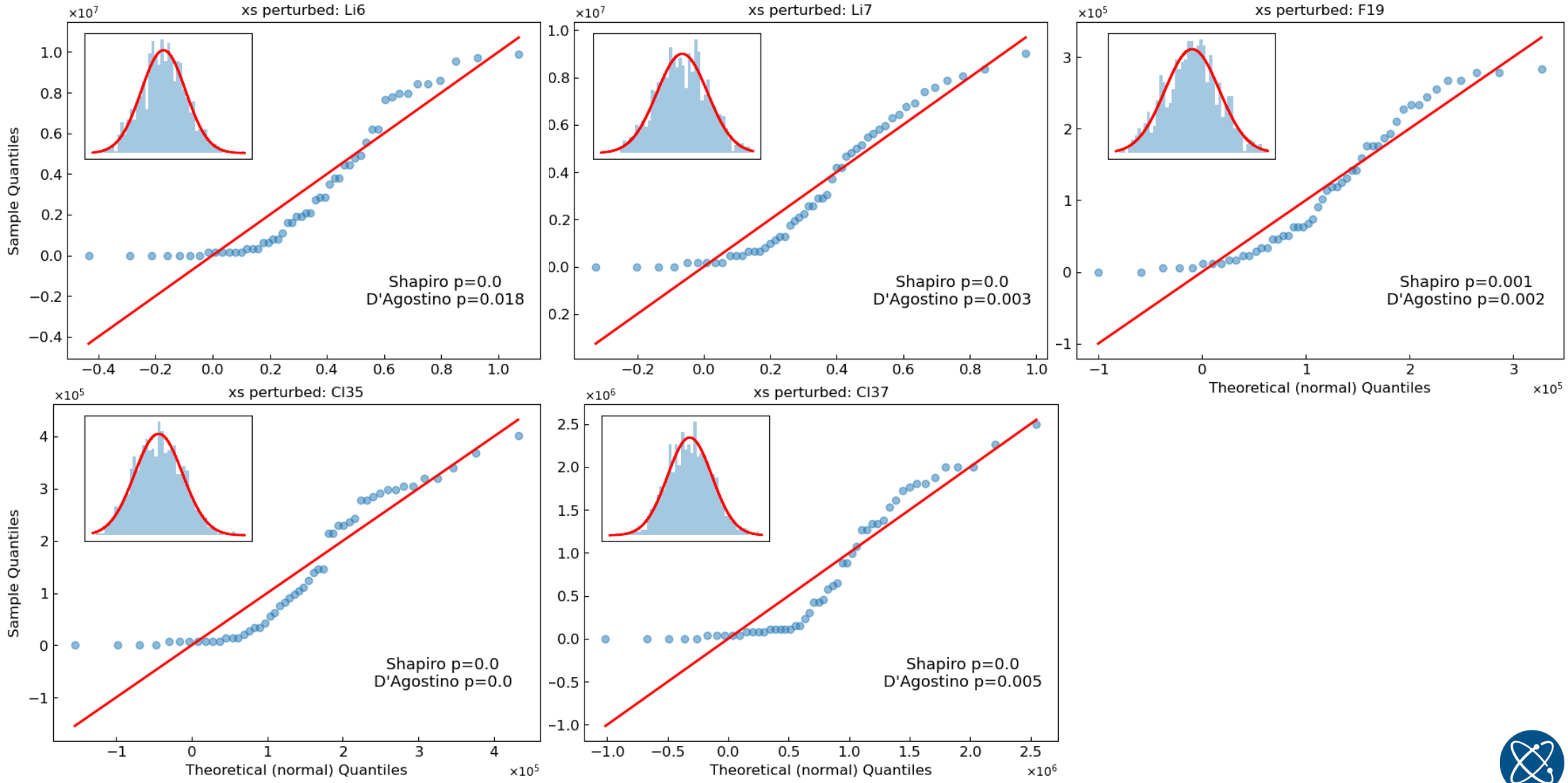


courtesy of B. Ebiwonjumi (MIT PSFC)

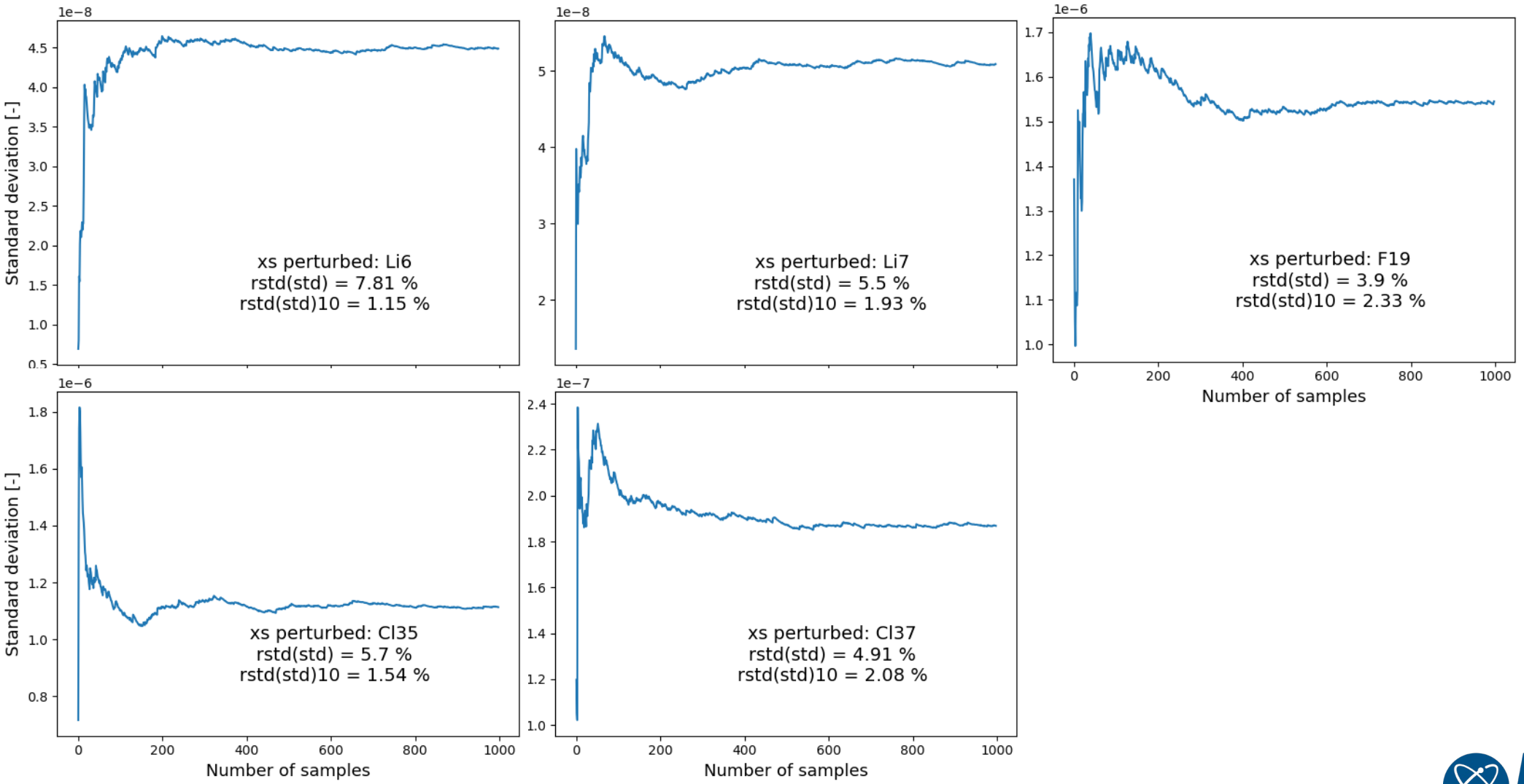
Actual distributions of computed TBR with xs uq



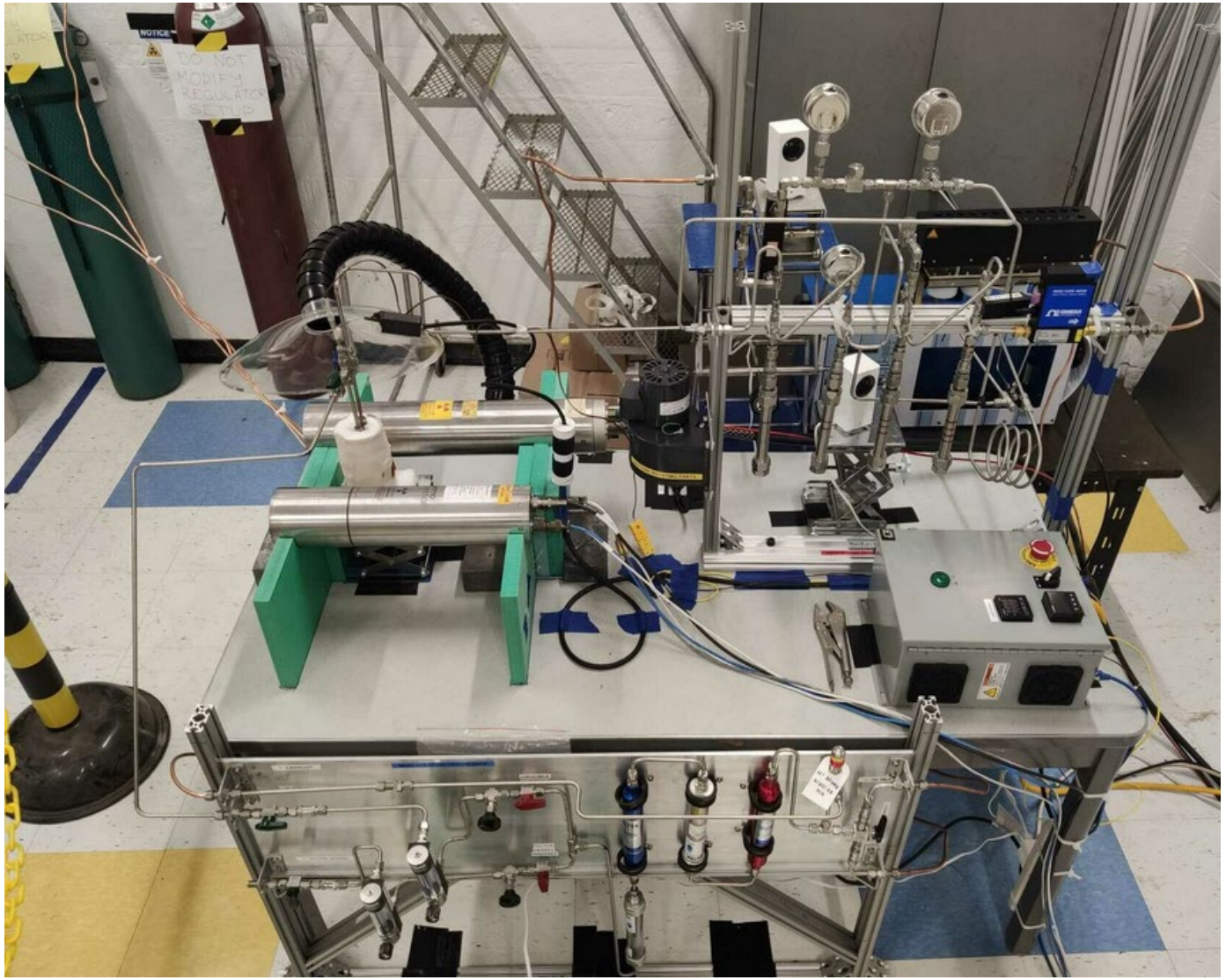
Actual distributions of computed TBR with xs uq (2)



Std. dev. well stabilized after 1000 samples per xs



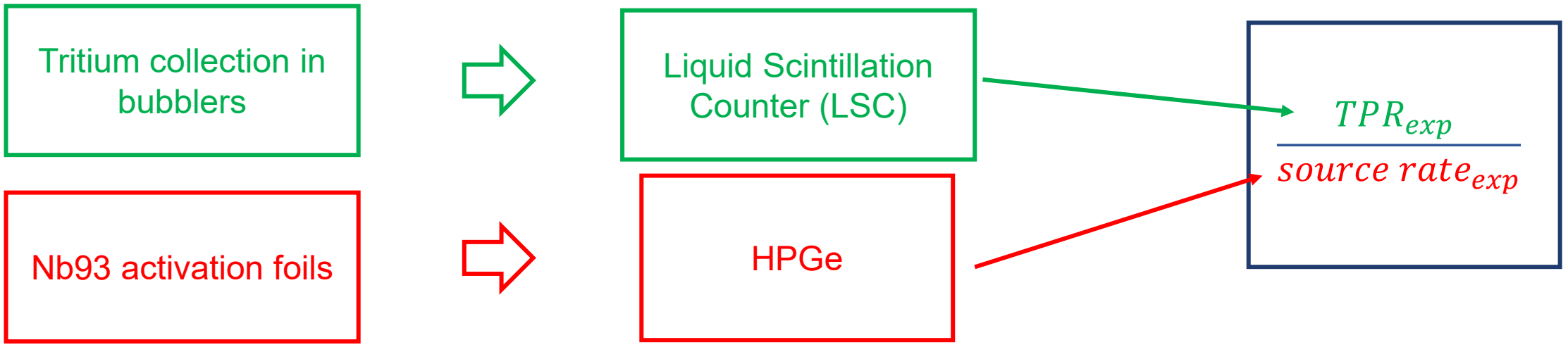
100 ml setup



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Measured TBR



More tritium transport

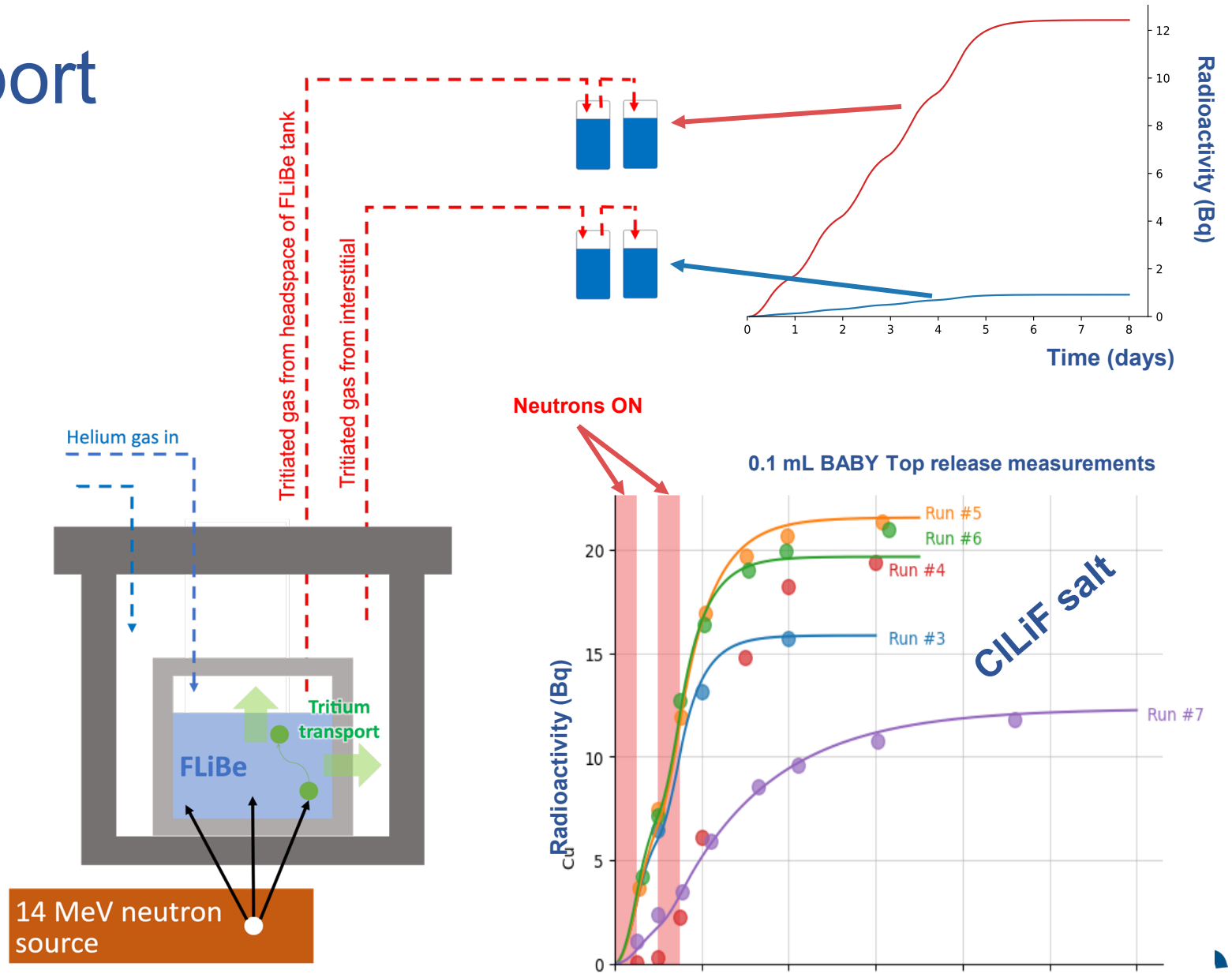
0D tritium transport model:

$$V\frac{\partial c}{\partial t} = S - A_{\mathrm{p}}\ k_{\mathrm{p}}\ c - A_{\mathrm{r}}\ k_{\mathrm{r}}\ c$$

V: salt volume
c: salt tritium concentration
S: tritium source
A_i: surface area
k_i: mass transport coefficient

Tritium water bubblers

- Bubblers collect 99% of the tritium each
- Water samples are analyzed with Liquid Scintillation Counting
- Measurements on BABY serve as validation data



More on chemistry and speciation

Species soluble in water (for bubblers):
- HTO

Insoluble:
- HT, TF

In run **#1** and **#2** we were expecting soluble forms already from salt but no tritium was collected
→ We added a **furnace** for insoluble → soluble collection
And we got (soluble) tritium!

In run #7 we added ~4% of EuF3 to ClLiF salt which changed the soluble-insoluble balance and, apparently also the salt release dynamics

